BLOOM · keys at home

`?` shows the keys on screen · `cmd` `Z` undoes anything · page 2: list modes, freezes, ideas

PLAY

`space`	play once (every list starts at its beginning)
`,`	loop (lists phase against each other)
`/`	pulsar: replay every few beats
`<` `>`	pulsar slower / faster
`.`	stop everything
`⇧` `space`	play the selected garden slot
`opt` `,`	loop / pulse the whole garden
`opt` `/`	
`~`	sustain pedal
`` ` ``	record / stop

NEW, UNDO, SAVE POINTS

`n`	new bloom in the same scale
`N`	new bloom, any of the twelve
`cmd` `z`	undo (last 300 changes)
`cmd` `⇧` `z`	redo
`opt` `p`	push a save point
`opt` `P`	pop back to it
`cmd` `S`	save / load the garden to a
`cmd` `0`	file (or the buttons top right)

GESTURES · whole bloom, small steps

`-`	**calm**: merge a note, soften, pitches inward
`=`	**intensify**: split a note, louder, pitches outward
`'`	**vary**: a little of everything moves
`{` `}`	merge two notes / split one in half

Rhythm only splits and merges, so the total length and your durations (halves and doubles) survive.

A bloom is five lists

pitch (a sub-list is a chord) · **time** to the next note, in beats · **velocity** (0 = rest) · **channel** · **intervals**. Each list can have its own length.

Phasing

Lists of different lengths cycle on their own: 5 pitches over 4 velocities shift the accent each time round. You hear it in the **loop** and **pulsar**; `space` always starts every list together. `L` shows a cursor per list.

WHOLE NOTES · each keeps its time, vel, chan

`s` `S`	scramble / perfect shuffle
`R`	mirror
`9` `0`	rotate ← →
`u` `U`	stutter / sputter
`d` `D`	braid / pyramid
`Y` `r`	spray / ratchet
`h` `H`	thin / thicken
`g` `G`	insert / remove a gap
`z`	drop last
`a` `A`	add a note: in the scale / any
`;` `:`	clump into chords / flatten chords
`W`	remove doubled notes
`opt` `w`	wheels within wheels: one accent drifting each cycle

`opt` + `s S R u U d D`: the same patterns on the pitches only.

PITCH · also in pitch mode

`1` `2`	semitone down / up
`!` `@`	octave down / up
`3` `4`	scale step down / up
`#` `$`	**breeze**: some notes a step down / up
`m` `M`	mutate: in the scale / chromatic
`i` `I`	invert / invert around the mean
`(` `)`	**pivot**: invert, keeping the highest / loudest note
`e` `E`	shear: some notes an octave out / center
`l` `L`	tilt down / up, steeper each press
`w`	draw a contour (in the scale)
`o` `O`	sort pitches up / down

SCALES AND KEYS · also in pitch mode

`k`	snap to the nearest named scale
`K`	**slant**: a neighbouring scale, any root
`[` `]`	one step **flatward / sharpward** (circle of fifths)
`j`	lift the garden slot's scale
`J`	reduce the scale
`T`	**tuning mode**: `]` `[` base, `9` `0` rotate, `3` `4` comma, `#` `$` breeze, `2` clear, `1` off/on
`opt-t`	**hold** the tuning over every bloom

Home or mode?

Home keys act on **whole notes** or the whole bloom; a mode applies the same verb to **one list**. `s` at home scrambles notes with their rhythm; `t` then `s` scrambles only the rhythm.

GARDEN · the row of slots

`↓`	save the bloom into the slot
`opt` `↓`	insert it here, the rest slide right
`↑`	lift the slot into the bloom
`←` `→`	move · with `⇧`, move and play
`↵` `⌫`	new slot / clear slot
`x` `X`	interlace / blend with the slot
`f`	take the slot's rhythm, dynamics, channels
`C`	cast: your pitch classes on the slot's contour and rhythm

MODES AND MENUS

`p` `t` `v` `c` `5`	pitch, time, velocity, channel, intervals mode (page 2)
`esc`	back home

FREEZE · `F`, then…

`p`	pitch: lock to the scale
`t` `r`	rhythm: the durations in use
`v`	dynamics: the levels in use
`c`	channels in use
`5`	intervals: the steps in use
`d`	total duration: trim → stretch → off
`g`	time grid
`e`	unpin all · **click** a node: pin · **drag** a box: keys change only those
`esc` `⇧` `esc`	clear / invert the box, else the pins

The menu stays open so you can set several; `F` or `esc` closes. The header shows what's frozen, e.g. ❄ **p t d:trim**.

VIEW

`opt` `1–9`	picture style
`opt` `c`	clean view for video (`esc` leaves)
`\`	console · `|` note labels
`opt` `\`	lanes: every list, each with its own cursor
`opt` `e`	edit the bloom as text
`cmd` `k`	command palette: every operation by name

INTERVALS · notes added at playback

`5`	**intervals mode** (randomize there with `n`)
`6` `^`	widen / narrow
`7` `&`	sparser / denser (a rest more / fewer)
`8`	reset (nothing added)
`5` `↵`	type them: `-P5/10`, `+3rd+5th`
`*`	bake into the notes as chords; a raga's fit too

BLOOM · list modes

p t v c 5 enter a mode · its key or esc goes home · another mode key switches across

In a mode every key acts on **that one list**; the others stay put. A key means the same verb in every mode. Play, garden, view, undo and F work as usual; anything else just shows a hint.

KEY	VERB	PITCH P	TIME T	VELOCITY V	CHANNELS C	INTERVALS 5
s S R 9 0 u U d D	scramble, shuffle, mirror, rotate, stutter, sputter, braid, pyramid: **the same in every list**. Changing one list's length makes it phase against the others.					
o O	sort up / down	low → high	short → long	crescendo	low → high	close → wide
z a A	drop last / add	add in scale / any	add one from the palette / at random			add a step in use / a fresh one
1 2	down / up	semitone	faster / slower ×1.2	softer / louder	channel ∓1	every interval ∓1
! @	big down / up	octave	double / half time	much softer / louder	—	an octave across (+7 ↔ −5)
3 4	one palette step	scale step	next duration you use	next level you use	next channel in use	next size in use
# $	breeze	as 3 4, but only some values move, and the same ones tend to keep moving				
− =	converge / diverge	narrower / wider range	evener / more uneven	flatter / wider dynamics	toward / away from the main channel	narrower / wider
y	unify	all the commonest pitch	all the commonest duration	all the average level	all the main channel	all the commonest step
i / I	invert / around the middle	high ↔ low / around the mean	long ↔ short	loud ↔ quiet	across the range / around the main	above ↔ below / wide ↔ close
l L	tilt down / up (builds)	tilt down / up	accel. / rit.	decresc. / cresc.	drift down / up	closing in / opening out
w	curve	contour	rubato	hairpins	smooth path	swell of widths
m M	mutate gently / wildly	scale steps / chromatic	from the palette / anything			
h H	fewer / more distinct values	one pitch class fewer / more	durations	levels	channels	kinds of step
q / Q	quantize / choose the setting	snap to the scale	to the grid / pick grid	to at most N levels / pick N		at most N kinds of step
n	new list	in scale · N any of 12	new times (from the palette)	new dynamics	new channels	new intervals
x / f	with the garden slot	lace this list with the slot's / copy the slot's				
Only in time mode	r **ratchet**: a few of the longest durations split into 2 or 3 equal parts (never the short ones)					
Only in pitch mode	every pitch key from home: ( ) pivot · [ ] flatward / sharpward · k K j J scales · e E shear / center · and 9 0 rotates the pitches alone					
Also in pitch & intervals mode	6 ^ 7 & 8 * mean what they mean at home; ↵ types the list (⇧↵ loops it to the notes)					

Palette

The distinct values a list already uses: your durations, dynamic levels, channels, interval steps. For pitch, the palette is the scale. "Step" keys move **along your own vocabulary**, so a groove of 1, ½, ¼ stays in halves and doubles.

How a freeze works

Switching one on takes a **snapshot** of what the list uses now. Anything outside it then **snaps to the nearest** allowed value: on every play, and after every change while a loop or pulsar runs.

Snapping rewrites the bloom; unfreezing doesn't undo it. Freeze again for a new snapshot.

Pins are per event, not per list: a pinned node keeps all its values. Sorts and rotations flow around it; a new bloom (n) grows around it.

Undo vs save points

cmd z steps back **one change at a time**. opt p marks a place you like; wander as far as you want and opt P jumps straight back. The **notice** top-left names each change for 5 s.

Breeze

$ move only **some** values a step. Notes that just moved are likelier to move again, so repeated presses drift a line rather than shuffle the whole thing.

Intervals

Each step adds a chord at playback, by name, **+ above, − below**: −P5, +3rd+5th (in the scale), · nothing. /N spaces it: −P5/10 is a fifth below every 10th note. The list cycles at its own length, so it drifts through the notes. * bakes it.

Moving through keys

K **slant** jumps to a near-neighbour scale on any root and avoids the last few it visited, so it wanders outward.

[ ] travel **steadily**: one scale note moves a semitone per press, C → G → D… sharpward or the reverse.

Pivot and pitch classes

( ) turn the melody upside down while the highest (or loudest) note stays put.

In pitch mode h H remove or add one **pitch class**; the scale follows.